**DKL Central Computer Company**

**KL Central Computer Company(KLCCC)** is a fast growing computer sales and repair company based in KL. Due to increasing volume of business transactions the company has decided to use an automated system.

Your team has been hired to provide a software solution written in Python to handle all transactions. There are 3 different functional areas:

Customer Management
User Management
Inventory Management

Each functional area MUST be managed by individual developer for ease of maintenance.

Login to the system
There will be two options:

- Sign up
- Login

The user must sign up to the system. If the new user is a customer, he/she will be approved by regular admin staff or Super user. If the new user is admin, he/she will be approved by the Super User.

Depending upon the type of user, the relevant menu with relevant functionality will be displayed.

All user login information will be held in a login data file.

Functionalities of Customer management:

- Registration via interface
  When a customer chooses to sign up, he/she must provide certain personal details to identify their uniqueness. Some of the information that can include IC/passport number, name, phone, city of domicile, date of registration etc.

- Login access to system
  Only customers approved by Admin Staff or Super User can access the system.

- Customer Menu
  - Purchase Order
    This option allows the customer to place orders for new computers/accessories etc. The list of available items is maintained by Inventory Management. No Purchase Order will be processed until full payment has been made

  - Service/Repair Order
    The user can Request for service or repair of a computer or a device using this option. The option will use information maintained by Inventory Management

to provide tentative timelines for service/repair. No Service/Repair Order will be processed till full payment has been made

- Modify Purchase/Service/Repair Order
  The user can choose to make changes to the Purchase Order/Service/Repair Order if no payment has been made.

- Payment for orders placed
  The user can use this to make payment. All orders must be paid before proceeding work.
- Inquiry of order status
  User can use this option to check status of their repair order
- Cancel Orders
  User can cancel any order BEFORE any payment has been made. Once payment is made, money is non-refundable.
- Reports
  This option allows user to view various reports of their options.

Functionalities of User Management

- Registration via interface
  Any new Admin user must also register using this interface just as customer and can only access once the super user has approved the user
- Login access to system
  Admin users need to be registered before they can use the login
- User Menu
  - Add Users
    Managed by Super user
  - Verify New Customers
    Managed by admin and super user
  - Modify User Personal Details
    Managed by super user.
  - Disable User Access (Super User Only)
    Only super user has access to this option.
  - Inquiry of User's system usage
    Accessible to super user only
  - Check Customer Order Status
    Available to all users
  - Reports
    Available to admin and super user only

Functionalities of Inventory Management

- Registration via interface
  Accessible to inventory staff only
- Login access to system
  Available only once inventory staff are approved

- Inventory Menu
    - Purchase Order for new computers/spare parts
      Used to place orders for new computers/parts needed for repair.
    - Stock Check/Adjustments
      Available only to inventory staff
    - Check Purchase Order status
      Available only to inventory staff/admin and super user
    - Modify Purchase Order
      Available to inventory staff only. It can only be done before the payment is made.
    - Cancel Purchase Order
      Unless the product is not available, all paid orders are non-refundable
    - Reports
      All reports related to inventory management are available here.

# 1.0 REQUIREMENTS

i. You are required to carry out extra research for your system and document any logical assumptions you made after the research.

ii. Your program should use symbolic constants where appropriate. Validations need to be included to ensure the accuracy of the system. State any assumptions that you make under each function.

iii. You are required to store all data in text files. There is no limit on the number of text files that can be used but they should be kept minimum.

iv. You are expected to use list and functions in your program. Your program must embrace modular programming technique and should be menu-driven.

v. You may include any extra features which you may feel relevant and that add value to the system.

vi. There should be no need for graphics in your program, as what is being assessed, is your programming skill not the interface design. The marking scheme for the assignment has been provided so that you clearly know how the assessment for this assignment would be done.

vii. You should include good programming practices such as comments, variable naming conventions and indentation**.**

viii. In a situation where a student:

- ***Failed to attempt the assignment demonstration, overall marks awarded for the assignment will be adjusted to 50% of the overall existing marks.***
- ***Found to be involved plagiarism, the offence and will be dealt in accordance to APU regulations on plagiarism.***

ix. You are required to use Python programming language to implement the solution. Use of any other language like C/C++/Java is not allowed.

x. Global variables, build in functions like min, max, sort, etc… are not allowed.

## 2.0 DELIVERABLES

You are required to submit a softcopy of:

i. Program coded in Python – submitted as .py file.
- Name the file under your name and TP number (e.g. KATHY_SIERRA_TP123456.py)
- Start the first two lines in your program by typing your name and TP number (e.g. as follows):
  #KATHY SIERRA
  #TP123456

ii. Text files created through test data – submitted as .txt files.

iii. A documentation of the system – submitted as NAME_TPNUMBER.pdf file - that incorporates basic documentation standards such as header and footer, page numbering and includes: - Cover page
- Table of contents
- Introduction and assumptions
- Design of the program – using pseudocode **and** flowcharts – which adheres to the requirements provided above
- Program source code with explanation
- Additional features source code with explanation
- Screenshots of sample input/output with explanation
- Conclusion

- References using Harvard Name Referencing

## 3.0 ASSESSMENT CRITERIA

i. Design (Pseudocode and Flowchart) 30%

Detailed, logical and accurate design of programmable solution.

ii. Coding / Implementation (Python code) 30%

Application of Python programming techniques (from basic to advance); good programming practices in implementing the solution as per design; and adequate validation meeting all system requirements with all possible additional features.

iii. Documentation 20%

Adherence to document standard format and structure; screen captures of input/output with explanation; and inclusion of generated text files.

iv. Demonstration 10%

Ability to run, trace code, explain work done and answer questions.

v. Q & A 10%

Ability to run, trace code, explain work done and answer questions.

## 4.0 PERFORMANCE CRITERIA

**Distinction (80% and above)**

This grade will be assigned to work which meets all of the requirements stated in the question. The program runs smoothly when executed. There is clear evidence and application of Python concepts up to advanced level. The program solution is unique with excellent coding styles and validation. The program implemented maps completely against the design (pseudocode and flowchart) as seen in the documentation. The design of the solution varies in styles and has unique logic with hardly any errors / omissions. The documentation does not have any missing components. Sample inputs/outputs documented have clear explanation. Student must be able to provide excellent explanation of the codes and work done, show additional concepts / new

ideas used in the solution, able to answer all questions posed with accurate / logical answers / explanation provided with sound arguments and clear discussion. Overall an excellent piece of work submitted.

**Credit (65%-74%)**

This grade will be assigned to work which is considered to be of good standard and meets most of the requirements stated in the question. The program runs smoothly when executed. There is clear evidence and application of Python concepts up to at least intermediate level. The program solution is unique with good coding styles and validation. The program implemented maps well against the design (pseudocode and flowchart) as seen in the documentation. The design of the solution varies in styles and has unique logic with minor errors / omissions. The documentation does not have any missing components. Sample inputs/outputs documented with some explanation. Student must be able to provide good explanation of the codes and work done, answer most questions posed with mostly accurate / logical answers / explanation. Overall a good assignment submitted.

**Pass (50%-64%)**

This grade will be assigned to work which meets at least half of the basic requirements (approximately 50%) stated in the questions. The program runs smoothly when executed. There is clear evidence and application of Python concepts at basic level. The program solution is common with basic coding styles and validation. The program implemented somewhat maps with the design (pseudocode and flowchart) as seen in the documentation. The design of the solution is average in terms of logic and style with some errors / omissions. The documentation has some missing components. Sample inputs/outputs documented but without any explanation. Student must be able to explain some codes and work done and able to answer some questions posed with some accurate / logical answers / explanation. Overall an average piece of work submitted.

**Fail (Below 50%)**

This grade will be assigned to work which achieved less than half of the requirements stated in the question. The program is able to compile but not able to execute or with major errors. The program solution has only basic coding styles with no validation. The program solution has little or no mapping with the design. The design of the solution has major / obvious errors / omissions. The documentation has some missing essential components. Student is barely able

to explain the codes / work done and answer given on the questions posed but with mostly inaccurate / illogical answers / explanation. Overall a poor piece of work submitted.